

Project Report

TITLE

Customer Segmentation and Revenue Optimization for Retail Company

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EXECUTIVE SUMMARY

This project analyzes retail transaction data to understand how customers behave and identify the most valuable ones. The goal is to help businesses increase revenue by targeting the right customers more effectively.

I performed RFM analysis to measure customer behavior based on recency, frequency, and monetary value. I then used clustering techniques such as K-Means and Hierarchical clustering in Python to group customers with similar buying patterns.

The optimal number of clusters was selected using the Elbow Method, and I evaluated cluster quality using the Silhouette Score.

Finally, I identified customer segments such as VIP, high-value, regular, and at-risk customers. These insights can be used to create targeted marketing campaigns, improve customer retention, and drive business growth.

PROBLEM DEFINITION

ABC Retail Pvt. Ltd. has been facing slow revenue growth despite increasing customer acquisition. A key issue is that all customers are treated the same, leading to ineffective marketing efforts and lower customer retention.

This project focuses on understanding customer purchasing behavior and identifying different types of customers based on their buying patterns. By segmenting customers into meaningful groups, the aim is to highlight high-value customers and those who may require targeted attention.

To achieve this, customer data was analysed using RFM (Recency, Frequency, Monetary) analysis, along with clustering techniques to group similar customers. These methods help uncover patterns in customer behavior and support better decision-making.

The results provide clear customer segments such as high-value, regular, and at-risk customers. These insights can help the company improve marketing strategies, increase repeat purchases, and drive overall revenue growth.

DATASET DESCRIPTION

The dataset consists of retail transaction records, where each row represents an individual purchase made by a customer. It captures important details about each transaction, helping us understand how customers interact with the business.

Features:

- **InvoiceNo** – Unique ID for each transaction
- **CustomerID** – Unique ID for each customer
- **Quantity** – Number of items purchased
- **UnitPrice** – Price of each item
- **InvoiceDate** – Date when the transaction happened
- **Country** – Location of the customer

This dataset helps us understand how customers shop over time.

It shows how often they buy and how much they spend.

We can also see changes in their buying habits.

This make it easier to understand overall customer behaviour.

METHODOLOGY

- The data was cleaned by removing missing values and invalid transactions to ensure accuracy
- New features were created, including TotalPrice and RFM metrics for better analysis
- RFM analysis was used to understand customer behavior:
 - Recency – how recently a customer made a purchase
 - Frequency – how often a customer buys
 - Monetary – how much a customer spends
- Log transformation was applied to reduce skewness and improve data distribution
- Standard scaling was used to prepare the data for clustering algorithms
- K-Means and Hierarchical clustering techniques were applied to group similar customers
- The Elbow Method was used to determine the optimal number of clusters
- Based on the analysis, four clusters were selected for segmentation
- The model performance was evaluated using the Silhouette Score to ensure good clustering quality

DATA PREPROCESSING

Before starting the analysis, data preprocessing was carried out to improve the quality and reliability of the dataset. Since real-world data often contains errors or missing values, cleaning the data was an important step.

First, records with missing Customer IDs were removed, as they could not be linked to any specific customer. Next, invalid transactions were filtered out by removing rows where the quantity or unit price was less than or equal to zero, as these values do not represent actual purchases.

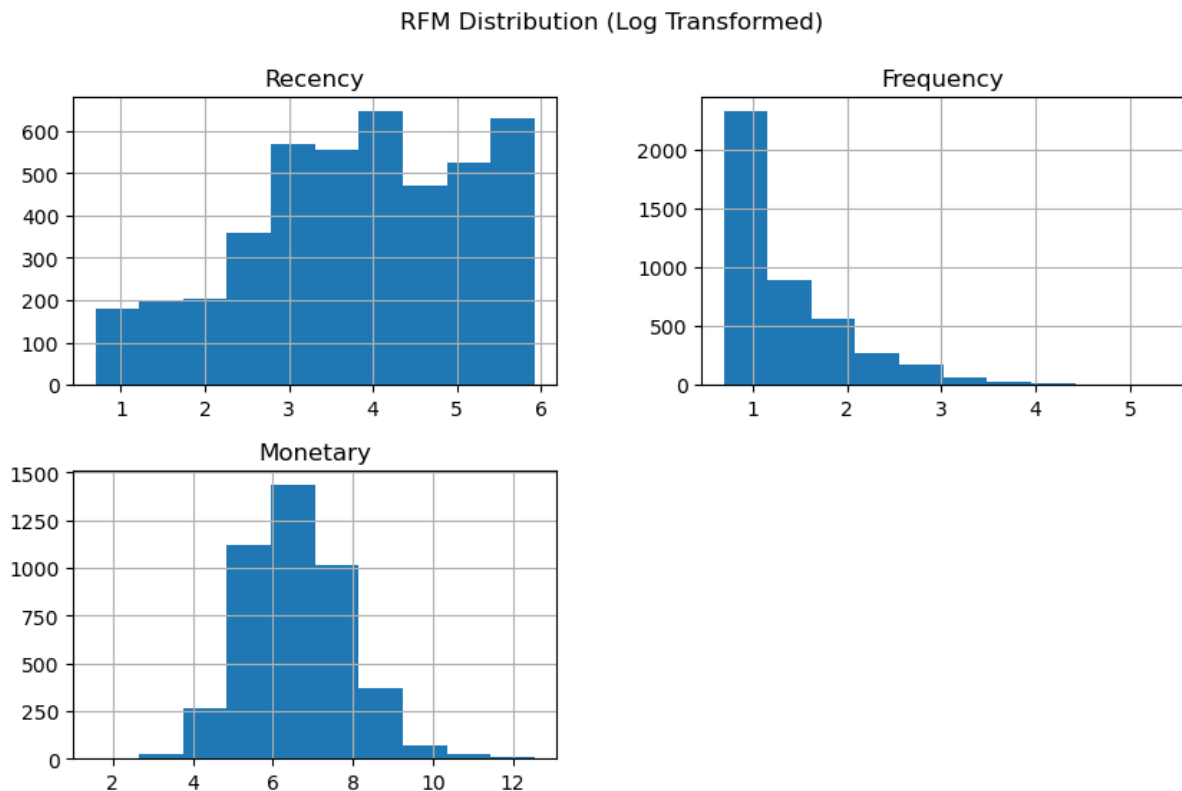
The InvoiceDate column was then converted into a proper date-time format to make it easier to analyse time-based patterns.

In addition to cleaning, some new features were created to make the analysis more meaningful. A new column called TotalPrice was calculated by multiplying Quantity and UnitPrice, which represents the total amount spent on each transaction. The month was also extracted from the InvoiceDate to study monthly trends in customer behavior.

These preprocessing steps helped transform the raw data into a clean and structured format, making it suitable for further analysis and modeling.

EXPLORATORY DATA ANALYSIS (EDA)

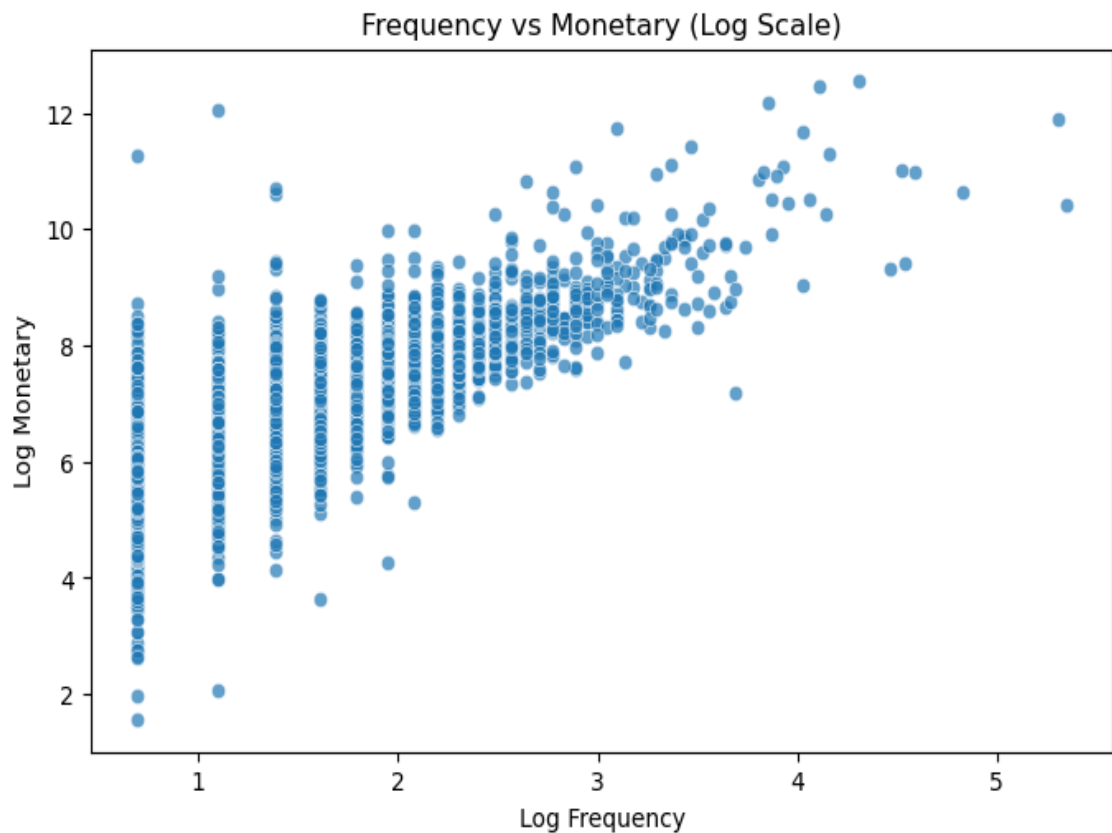
Univariate Analysis



Observations:

- Most customers do not purchase frequently
- A small group of customers contributes a large portion of the revenue
- The data is not evenly distributed and shows a skewed pattern
- Customer spending varies significantly across different groups

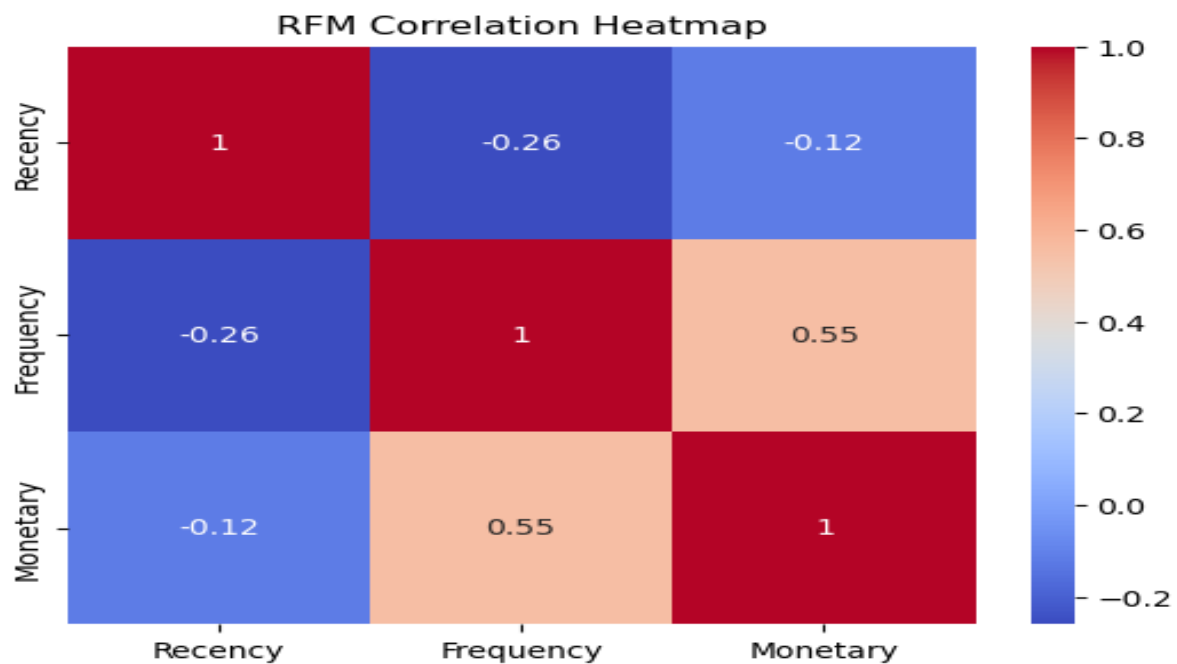
Bivariate Analysis



Observations:

- There is a positive link between how often customers buy and how much they spend
- Customers who buy more often usually spend more
- Higher frequency customers tend to generate more revenue
- There is a clear upward trend in the data

Correlation Analysis



Findings:

There is a clear relationship between how often customers buy and how much they spend. Customers who purchase more frequently tend to spend more money.

On the other hand, recency shows an opposite trend. Customers who have purchased recently are more active, while those who have not purchased for a long time are less engaged.

This indicates:

- Customers who bought recently are more likely to stay active
- Customers who buy more often usually spend more
- Customers who haven't purchased in a while may be at risk of leaving

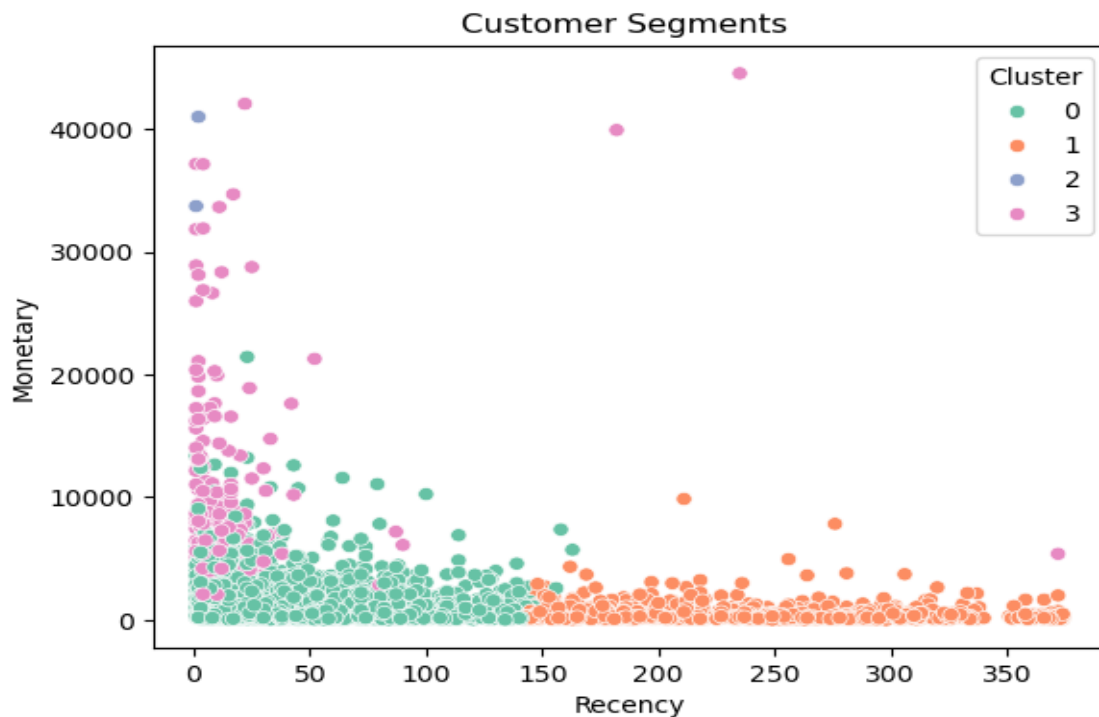
MODEL BUILDING

Clustering helps divide customers into groups based on similar behavior.

K-Means Clustering

- K-Means was used to group similar customers
- 4 clusters were selected based on analysis
- Each cluster represents a different customer type
- Helps in separating customers based on their buying behavior

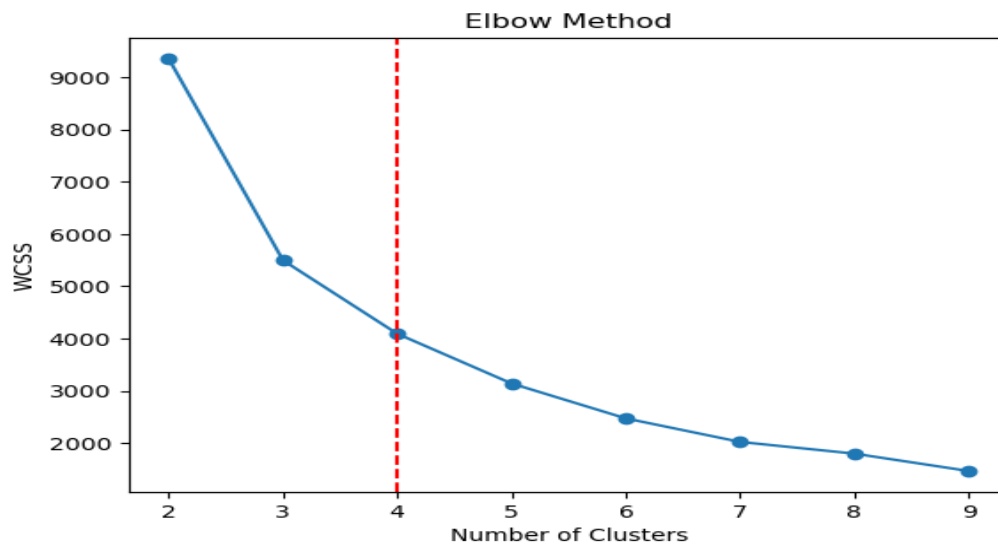
The following visualization shows customer segments formed using K-Means clustering.



Customer segmentation using K-Means clustering (Recency vs Monetary)

Elbow Method

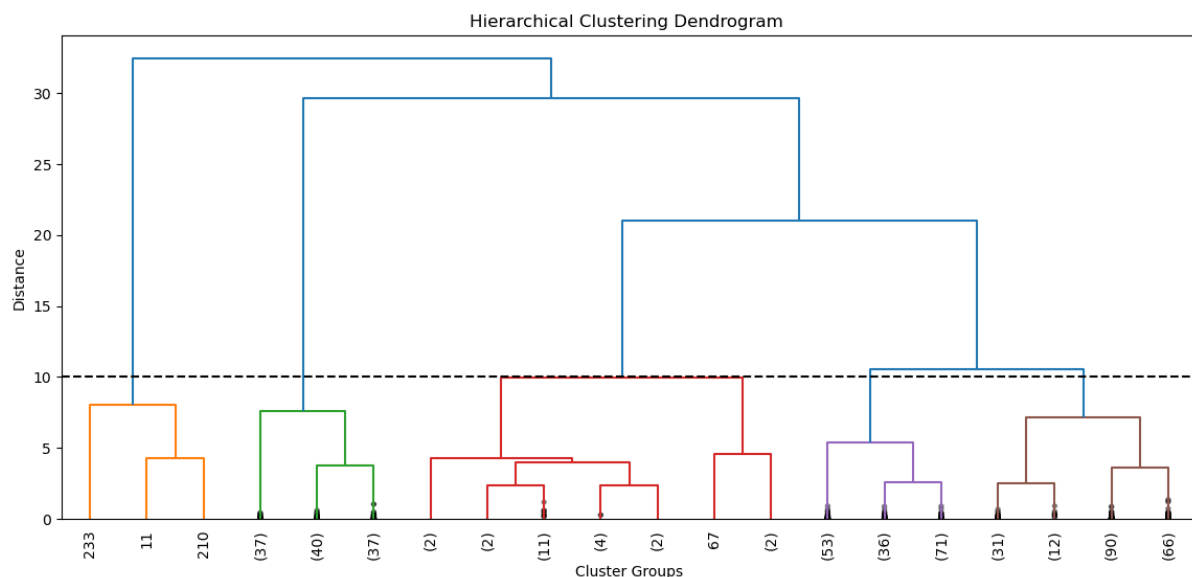
- Used to find the best number of clusters



Observation:

- The value drops quickly up to 4 clusters
- After that, the change becomes small

Hierarchical Clustering



Hierarchical clustering dendrogram showing cluster formation.

Model Comparison

Both K-Means and Hierarchical clustering methods were used to group customers based on their behavior.

K-Means produced well-separated clusters with a silhouette score of 0.61, indicating effective segmentation of customers.. It was also faster and worked efficiently with a large dataset, making it more suitable for this project.

Hierarchical clustering was useful for understanding how different customer groups are related to each other through a tree-like structure. However, it was slower and not as efficient when working with a large amount of data.

Based on these observations, K-Means was selected as the final model because it provided clearer segmentation and better performance.

EVALUATION

Clustering performance was evaluated using the Silhouette Score.

Result:

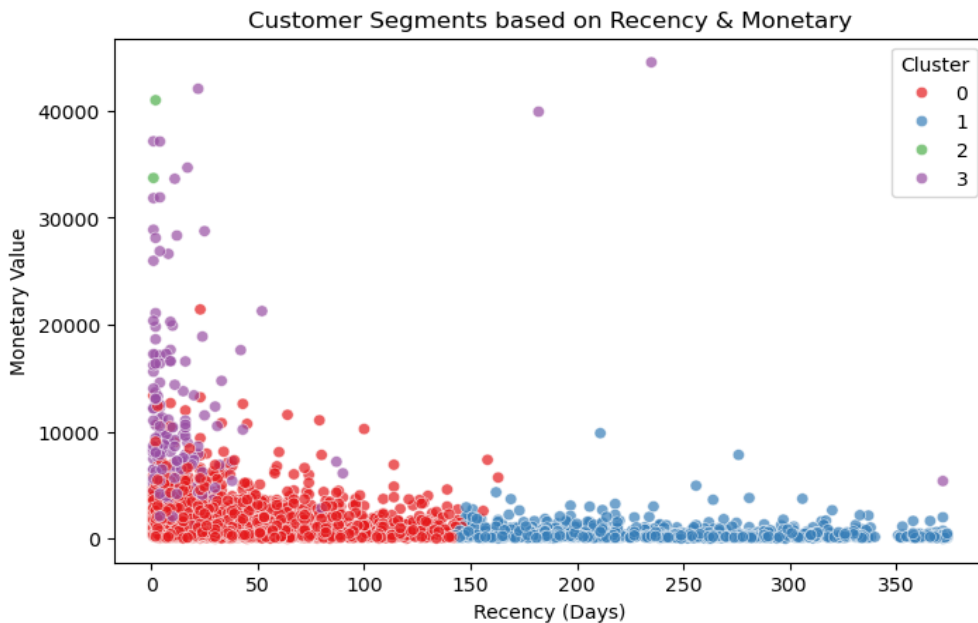
- Score ≈ 0.61

Interpretation:

- The clusters are well separated
- Customers within each group are similar to each other
- The segmentation is meaningful and reliable

RESULTS & INSIGHTS.

The following visualization shows the final customer segments based on Recency and Monetary values.



Final customer segmentation based on Recency and Monetary values.

CLUSTER PROFILING TABLE

	Recency	Frequency	Monetary
Cluster			
0	43.702685	3.682711	1359.049284
1	248.075914	1.552015	480.617480
2	7.384615	82.538462	127338.313846
3	15.500000	22.333333	12709.090490

Customer Segments

Cluster 0 – Regular Customers

- These customers show average activity
- They do not purchase very often
- Their spending is moderate

Cluster 1 – At-Risk Customers

- These customers have not purchased recently
- They buy very rarely
- Their spending is low

Cluster 2 – VIP Customers

- These customers purchased very recently
- They buy very frequently
- They spend a lot and are the most valuable group

Cluster 3 – High-Value Customers

- These customers are active
- They purchase regularly
- Their spending is high

Key Insights

- A small group of customers generates most of the revenue
- VIP customers are very important for the business
- Some customers may stop buying if not targeted properly
- Customer behavior is different across each group

CONCLUSION

- Customer segmentation was done using RFM analysis and clustering techniques
- Customers were grouped based on their purchasing behavior
- A small number of customers contribute most of the revenue
- Some customers are less active and may stop buying
- The analysis helps in understanding customer patterns better
- It supports better decisions and improves customer retention
- Overall, it helps the business grow and perform better

RECOMMENDATIONS

- Give special rewards and offers to VIP and high-value customers to keep them loyal
- Try to bring back inactive customers using discounts and marketing campaigns
- Encourage regular customers to buy more often with combo offers and limited-time deals
- Focus more on high-value customers to increase revenue and get better results

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